

NOTES ON A SCHEMA FOR STORIES

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I. INTRODUCTION

Just as simple sentences can be said to have an internal structure, so too can stories be said to have an internal structure. This is so in spite of the fact that no one has ever been able to specify a general structure for stories that will distinguish the strings of sentences which form stories from strings which do not. Nevertheless, the notion of "well-formedness" is nearly as reasonable for stories as it is for sentences. Consider the following examples:

(1) Margie was holding tightly to the string of her beautiful new balloon. Suddenly, a gust of wind caught it. The wind carried it into a tree. The balloon hit a branch and burst. Margie cried and cried.

(2) Margie cried and cried. The balloon hit a branch and burst. The wind carried it into a tree. Suddenly a gust of wind caught it. Margie was holding tightly to the string of her beautiful new balloon.

Here we find two strings of sentences. One, however, also seems to form a sensible whole, whereas the other seems to be analyzable into little more than a string of sentences. These examples should make clear that some higher level of organization takes place in stories that does not take place in strings of sentences. The purpose of this chapter is to illustrate that point, to develop some notions of the sorts of structures that might be involved, and to illustrate how these structures can be used to produce cogent summaries of stories.

To begin, it is clear that simple sentences are not the highest level of structured linguistic input. Sentences themselves can serve as arguments for higher predicates and thus form more complex sentences. For example,

(3) Margie knew that her balloon had burst.

Here we have one sentence about the bursting of Margie's balloon embedded as the argument of a higher verb. Sentences such as these, of course, occur with high frequency. Another case in which sentences occur as arguments of higher predicates is

(4) Margie cried and cried because her balloon broke.

In this case the predicate "because" takes two sentences as arguments. Now consider the following pair of sentences:

(5a) Margie's balloon broke.

(5b) Margie cried and cried.

It seems clear that the sentence pair (5a) and (5b) have almost the same meaning as (4) and ought therefore to have the same underlying structure. Thus if we are to understand correctly (5a) and (5b) we must infer the causal relationship between the propositions. This, I suspect, is but a scratch on the surface of the kinds of "supra-

sentential" relationships that are implied and understood in ordinary discourse. In particular, I suggest that the structure of stories is ordinarily more than pairwise relationships among sentences. Rather, strings of sentences combine into psychological wholes. In the following section I explore the nature of these wholes and propose a simple story grammar which accounts for many of the salient facts about the structure of simple stories and which will serve as the basis for a theory of summarization.

II. A SIMPLE STORY GRAMMAR

A. The Grammar Rules

In this section I will develop a grammar which I suggest accounts in a reasonable way for the structure of a wide range of simple stories. The grammar consists of a set of syntactical rules which generate the constituent structure of stories and a corresponding set of semantic interpretation rules which determine the semantic representation of the story. The symbol "+" is used to form two items in a sequence; the symbol "|" is used to separate mutually exclusive alternatives. A "*" following a structure name indicates one or more of those units; for example, A* is one or more As.

Rule 1: Story -> Setting + Episode

The first rule of our grammar says simply that stories consist of a Setting followed by an Episode. The Setting is a statement of the time and place of a story as well as an introduction to its main characters. The Setting corresponds to the initial section of stories such as:

Once upon a time, in a far away land, there lived a good king, his beautiful queen, and their daughter Princess Cordelia . . .

The setting is usually just a series of stative propositions, often terminated by phrases such as:

One day, as Princess Cordelia was walking near the palace . . .

In the story illustrated in the first example--the Margie story--the setting consisted of the sentence:

Margie was holding the string of her beautiful new balloon.

The remainder of the story is an Episode. The simple semantic rule corresponding to Rule 1 is:

Rule 1': ALLOW (Setting, Episode)

Semantically, the setting forms a structure into which the remainder of the story can be linked. It plays no integral part in the body of the story and under certain conditions can be eliminated without adversely effecting the story. In such cases, the characters and their relevant characteristics must be introduced in the body of the story.

Rule 2: Setting -> (State)*

Rule 2 simply expresses the assumption that settings consist of a set of stative propositions.

Rule 2': AND (State, State, . . .)

Semantically, the states are represented as a set of conjoined propositions entered into the data base.

The first real substantive rule in our rewrite for Episode is:

Rule 3: Episode -> Event + Reaction

Episodes are special kinds of events which involve the reactions of animate (or anthropomorphized) objects to events in the world. The episode consists merely of the occurrence of some event followed by the reaction of the hero of the episode to the event. Our semantic rule corresponding to Rule 3 is:

Rule 3': INITIATE (Event, Reaction)

That is, the relationship between the external event and the hero's reaction is one that I call INITIATE. I have taken the term from Schank (1973), although my use is slightly different from his. I use the term INITIATE to represent a kind of causal relationship between an external event and the willful reaction of a thinking being to that event. In the Margie story illustrated in (1), I assume that the relationship between Margie's crying and the breaking of her balloon is the INITIATE relationship. Presumably, the crying is mediated by an internal mental response such as "sadness".

Event is the most general category of our entire grammar. The following rule expresses the structure of an event:

Rule 4: Event -> {Episode | Change-of-state | Action | Event + Event}

Thus an Event can be any of the alternatives, an episode, a simple change of state, or an action that people carry out. All are special kinds of events. Furthermore, a sequence of events also can constitute an event. The first three parts of Rule 4 require no semantic interpretation rules. Our semantic rule corresponding to the fourth rewrite for event is:

Rule 4': CAUSE (Event, Event) or ALLOW (Event, Event)

The rule states that a sequence of two events can either be interpreted as one event CAUSE a second event or they can be interpreted as the first event ALLOW the second. The term CAUSE is used when the relationship between the events is one of physical causation as in the balloon hitting the branch causing the balloon to break in the Margie story. (The CAUSE predicate is similar to Schank's (1973) RESULT.) ALLOW is a relationship between two events in which the first makes the second possible, but does not cause it; thus the relationship between the wind catching the balloon and the wind carrying it into the tree I would say is ALLOW. (Here again my usage of ALLOW is clearly closely related to Schank's ENABLE, but is probably not identical.)

Rule 5: Reaction -> Internal Response + Overt Response

Thus a reaction consists of two parts, an internal and an overt response. The semantic relation between these two responses is:

Rule 5': MOTIVATE (Internal Response, Overt Response)

MOTIVATE is the term used to relate thoughts to their corresponding overt actions.

Presumably there are a large variety of types of internal responses. The two most common, however, seem to be emotions and desires. Thus we have:

Rule 6: Internal Response -> {Emotion | Desire}

Presumably other internal responses can be aroused, but in the stories I have analyzed these two have been sufficient. The overt response is, of course, semantically constrained to be a plausible response for our particular internal response.

B. An Example

Before continuing the explication of the grammatical rules, it may be useful to illustrate the structure--semantic and syntactic--generated by these rules for the Margie story. Fig. 1 illustrates the analysis. Fig. 1a gives the units of the Margie story. Fig. 1b illustrates the tree generated by application of the syntactic rules. Fig. 1c illustrates our semantic structures.

- (1) Margie was holding tightly to the string of her beautiful new balloon.
- (2) Suddenly, a gust of wind caught it
- (3) and carried it into a tree
- (4) It hit a branch
- (5) and burst.
- (6) [sadness]
- (7) Margie cried and cried.

Fig. 1a. The units of the Margie story.

In the semantic structures the convention is followed that predicate names written in the ovals and the arguments of the predicates are pointed to by arrows. The propositions which are the units of the story are simply numbered. Note also that inferred propositions are written in the body of the story in square brackets. This simply facilitates our reference to these propositions. In the semantic structures, I follow the convention that the left-hand arrow points to the "causal event" and the right-hand arrow points to the "caused event". This notation is similar to that given in Rumelhart, Lindsay, & Norman (1972) and in Norman, Rumelhart, & the LNR research group (1975).

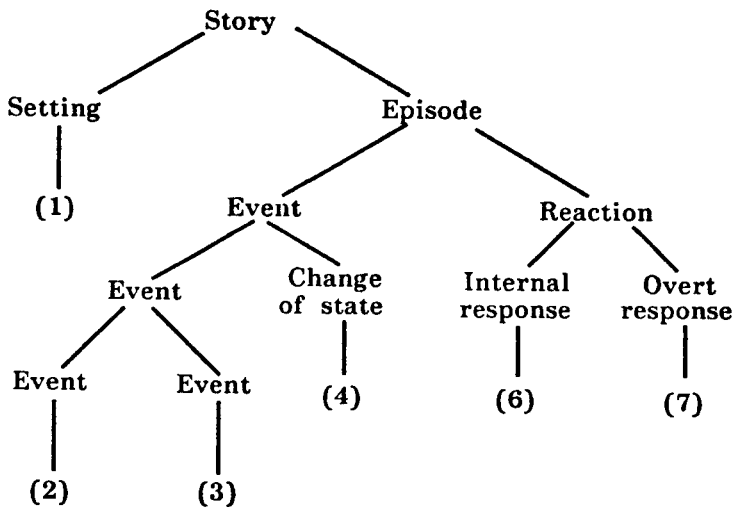


Fig. 1b. The syntactic structure of the story.

I will return, in a later section of this chapter, to a discussion of the role of our semantic and syntactic structures in story processing. At this point, however, we return to the development of our story grammar.

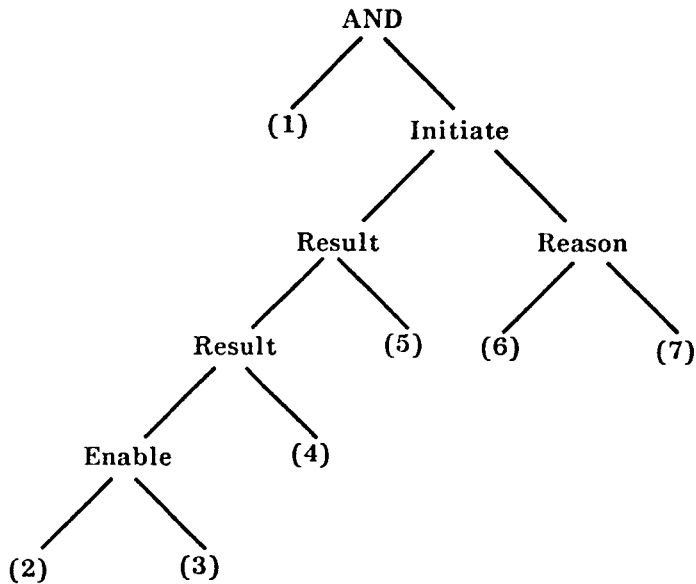


Fig. 1c. The semantic structure of the story.

In the case illustrated above, the overt response node had a trivial expansion--namely it was interpreted simply as an action. Other times it has a more complex expansion. In particular, when the internal response is a desire, the overt response is an attempt or rather a series of attempts at obtaining that desire. Thus we have the following syntactic rules:

Rule 7: Overt Response $\rightarrow$ {Action | (Attempt)*}

Corresponding to Rule 7 we have the following semantic interpretation rule:

Rule 7': THEN (Attempt, Attempt, . . .)

The THEN relation is a predicate of any number of arguments of which the n th argument occurs prior to the n -1st argument in time.

Now, attempts too have their own internal structure:

Rule 8: Attempt $\rightarrow$ Plan + Application

TABLE I.
Syntactic Rules and Semantic Interpretation Rules

-
- (1) Story -> Setting + Episode
=> ALLOW (Setting, Episode)
 - (2) Setting -> (States)*
=> AND (State, state,.....)
 - (3) Episode -> Event + Reaction
=> INITIATE (Event, Reaction)
 - (4) Event -> {Episode | Change-of-state | Action | Event + Event}
=> CAUSE (Event₁, Event₂) or ALLOW (Event₁, Event₂)
 - (5) Reaction -> Internal Response + Overt Response
=> MOTIVATE (Interval-response, Overt Response)
 - (6) Internal Response -> {Emotion | Desire}
 - (7) Overt Response -> {Action | (Attempt)*}
=> THEN (Attempt₁, Attempt₂,.....)
 - (8) Attempt -> Plan + Application
=> MOTIVATE (Plan, Application)
 - (9) Application -> (Preaction)* + Action + Consequence
=> ALLOW (AND(Preaction, Preaction,...),
{CAUSE | INITIATE | ALLOW} (Action, Consequence))
 - (10) Preaction -> Subgoal + (Attempt)*
=> MOTIVATE [Subgoal, THEN (Attempt,.....)]
 - (11) Consequence -> {Reaction | Event}
-

In other words, an attempt consists of two parts, the development of a plan (to obtain the desire) and the application of that plan, Rule 8' is the semantic interpretation for Rule 8:

Rule 8': MOTIVATE (Plan, Application)

In words, Rule 8' states that the Plan MOTIVATES the application of the plan. The application consists of three parts; an optional series of preactions optionally followed by an action, and a consequence of that action. Thus we have:

Rule 9: Application -> (Preaction)* + (Action + consequence)

TABLE II.
Definitions of Semantic Relationships

AND	A simple conjunctive predicate of any number of arguments.
ALLOW	The relationship between an event which made possible, but which did not directly cause a second event. Thus for example, going to the store ALLOWS but does not cause me to buy some bread.
INITIATE	The relationship between an external event and the willful reaction of an anthropomorphized being to that event. Thus for example, an angry lion escaping from a cage in front of me INITIATES my being afraid and running for safety.
MOTIVATE	The relationship between an internal response and the actions resulting from that internal response. Thus for example, my being afraid MOTIVATES my running to safety.
CAUSE	The relationship between two events in which the first is the physical cause of the second. Thus for example, the baseball striking the bat CAUSES the baseball flying towards the outfield.
THEN	The relationship which holds among a temporally sequenced set of events. The first argument occurs prior in time to the second, the second prior to the third, etc.

The semantic interpretation of this rule is:

Rule 9': **ALLOW** [**AND** (Preaction, . . .) (**INITIATE** | **CAUSE** | **ALLOW**) (Action,Consequence)]

In words, Rule 9' states that the temporally sequenced set of preactions **ALLOW** the action which either **INITIATES**, **CAUSES** or **ALLOWS** the Consequence. Which of these three semantic relationships are appropriate

depends on the nature of the action and its consequence. If none of the preactions are present, the semantics of the embedded proposition become the interpretation for the entire rule. Furthermore, in some stories one or more of the preactions fail and thus the action itself may never appear. In such cases the consequence also fails to appear and the semantics of Rule 9' is simply the temporal sequence of those preactions which do occur.

Two rules remain to complete the development of the story grammar. These are the rules for rewriting preactions and for rewriting consequences:

Rule 10: Preaction -> Subgoal + (Attempt)*

Rule 10': MOTIVATE [Subgoal, THEN (Attempt, . . .)]

Rules 10 and 10' state that preactions consist of the development of a subgoal which serves to MOTIVATE a series of attempts to obtain that subgoal. The consequence is a very general category. Consequences, it seems, can be any sort of event resulting from the action. Thus we have:

Rule 11: Consequence -> {Reaction | Event}

Before turning to detailed examples in which this grammar has been applied to stories, it will be useful to review the various rules and terms of the grammar. Table I summarizes the entire set of syntactic and semantic rules. Table II consists of my definitions of the semantic relations used in the statement of the semantic interpretation rules and Table III gives verbal descriptions of the syntactic categories used in the formulation of the syntactic rules.

III. ANALYZING A STORY

I now turn to an example of how the grammar is to be used to discover the structure underlying a simple story. As an example, I will begin with the analysis of one of Aesop's Fables entitled "The Man and the Serpent". The text of the story is given in Table IV. It should be

TABLE III.
Definitions of Syntactic Categories

Action	An activity engaged in by an animate being or a natural force.
Application	The process of attempting to carry out some plan for obtaining a desire.
Attempt	The formulation of a plan and application of that plan for obtaining a desire.
Change-of-state	An event consisting only of some object changing from one state to another.
Consequence	The outcome of performing an action for the purpose of obtaining some particular state of the world.
Desire	An internal response in which one wants and will thus probably try to obtain some particular state of the world.
Emotion	An internal response which consists of the expression of feelings.
Episode	An event and the reaction of an animate being to that event.
Event	A change of state or action or the causing of a change of state or action.
Internal Response	The mental response of an animate being to an external event.
Overt Response	The willful reaction of a willful being to an internal response.
Plan	The creating of a subgoal which if achieved will accomplish a desired end.
Preaction	An activity which must be carried out in order to enable one to carry out a planned action.
Reaction	The response of a willful being to a prior event.
Setting	The introduction to the characters and conditions of the characters in a story.
State	A property or condition of an object or a stable relationship among a set of objects.
Story	A kind of structured discourse which centers around the reactions of one or more protagonists to events in the story.
Subgoal	A goal developed in service of a higher more central goal.

TABLE IV.
The Text of "The Man and the Serpent"

-
- (1) The Man and the Serpent
 - (2) A countryman's son, by accident, trod upon a serpent's tail.
 - (3) The serpent turned
 - (4) and bit him.
 - (5) He died.
 - (6) The father, in revenge,
 - (7) got his axe,
 - (8) pursued the serpent,
 - (9) and cut off part of his tail.
 - (10) So the serpent, in revenge,
 - (11) began stinging several of the farmer's cattle.
 - (12) This caused the farmer severe loss.
 - (13) Well, the farmer thought it best to make it up with the serpent.
 - (14) So he brought food and honey to the mouth of its lair,
 - (15) and said to it "Let's forget and forgive; perhaps you were right in trying to punish my son, and take vengeance on my cattle, but surely I was right to revenge him; now that we are both satisfied why should we not be friends again?"
 - (16) "No, no," said the serpent; "take away your gifts; you can never forget the death of your son, nor I the loss of my tail."
-

noted that the text is already parsed into lines containing single propositions. These lines will be treated as the units on which the grammar operates. This fable is somewhat unusual in that there is virtually no setting for the story. The introduction of characters is completed in the title and the story proper starts right out with an Episode. As we shall see shortly, this story is properly analyzed as a sequence of embedded episodes. To illustrate the analysis, I will begin with the first episode of the story. The first episode consists of lines (2)-(5) of the story. The syntactic structure of this episode is given in Fig. 2a and the semantic structure in Fig. 2b.

It should be noted especially that the internal response and the Plan development both had to be inferred from succeeding actions. As will be noted in our discussions of summaries in the following section, it is very frequent that the nature of the mental events are not expressed when compressing the description of an event. Fig. 2b shows the proposed semantic structure of this episode; the structure is a completely right branching set of embedded causatives. As we shall also discuss in the following section structures such as these can be readily summarized by, for example, simply stating the most deeply embedded causal structure. In this case the structure (4) results in (5).

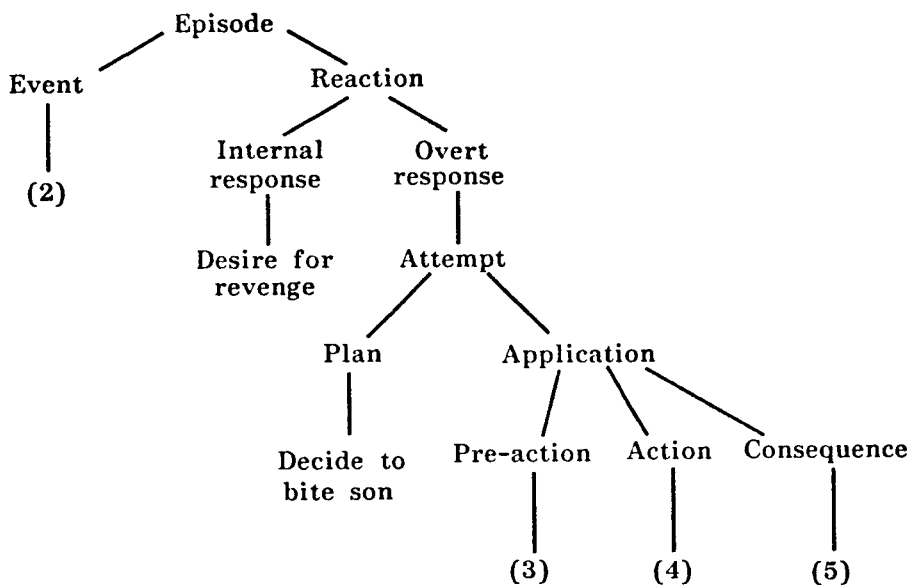


Fig. 2a. Syntactic structure of story.

Now, the next episode of the story actually includes this first episode as the initiating event and lines (6) - (9) of the story simply give us the farmer's Reaction to this event. The structure of the second episode is essentially the same as that for the first episode with the exception that the Internal Response--the desire to get revenge--is

explicitly mentioned in the text. The second interesting aspect of the structure is that it seems that the action and the consequence are lexicalized together into the verb "cut off". This suggests that our grammar should really work on the level of decomposed predicates rather than surface propositions. On the other hand, the loss is not so great working with surface propositions and the process of applying the grammar is really much simplified.

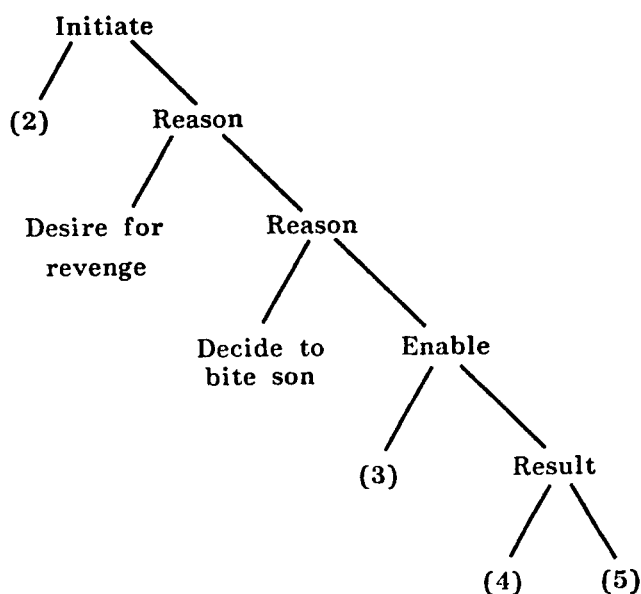


Fig. 2b. Semantic structure of story.

No new problems are introduced in the analysis of the remaining two episodes of the story. In each case, the prior episode serves as the initiating event for the immediately following episode.

In addition to this and the Margie story discussed above, the grammar has been applied in detail to four additional stories and applied informally to many others. The grammar, as described above, is adequate for all of the

stories analyzed in detail and it is my impression that most folk tales, fables, and the like can be analyzed without many new problems arising.

Rather than continue with a detailed analysis of the grammar and how various stories can be analyzed in terms of it, I now turn to a discussion of what one can do with a grammar of the form outlined above. In the following two sections I discuss two distinct ways in which this grammar can be used to extend our understanding of the appropriate linguistic analysis of connected discourse.

IV. SUMMARIZING STORIES

Connected discourse differs from an unrelated string of sentences (among other ways) in that it is possible to pick out what is important in connected discourse and summarize it without seriously altering the meaning of the discourse. The same is not true of strings of unrelated sentences. Since such strings lack structure and do not make meaningful wholes they cannot be summarized at all.

In this section a set of rules are developed which map our semantic structures onto summaries of these structures. In order to evaluate the summary rules, eight subjects were asked to read each of six stories including the "Margie" story and "The Man and the Serpent". The subjects were asked to give brief summaries of each of the stories. [Subjects had copies of the stories in front of them while performing this task.] In the remainder of this section I will present and discuss the summarization rules developed and show how they relate to the summarization responses actually observed.

A. The Summarization Rules

Summarization rules are defined on each of the sorts of semantic structures that can be generated by our grammar. Thus there is one or more summarization rule for each of the substantive semantic relationships defined in Table II.

B. Summarizing CAUSES

The relationship CAUSE[Event1,Event2] is summarized by according one of four different rules for summarizing such events. We will illustrate this with a set of examples. Suppose, the following sentence is encountered:

- (6) The snake bit the countryman's son causing him to die.

Here the CAUSE relationship holds between the event of the snake biting the boy and the event of his death. The following summaries are typical:

- (7a) The snake caused the countryman's son to die.

or in a more naturally lexicalized formulation

- (7b) The snake killed the countryman's son.

Here we have summarized the actual causal event by mere mention of the agentive force in that causal event. Thus the following summarization rule:

S1: Summary(CAUSE[X,Y])→Agent(X) caused (Y)

It is frequently the case that the "cause" and the resultant (Y in this case) are lexicalized into one term (e.g., cause to die here becomes kill).

A second summarization rule applicable to the CAUSE relationship is much like the first; it involves the use of *instrument* rather than agent. Thus if we encounter the following sequence:

- (8) The wind carried the balloon into the branch of a tree and it broke.

(8) would be interpreted as involving a CAUSE relationship between the two events in question. By rule S1, this can already be summarized as

- (9) The wind caused the balloon to break.

or

(10) The wind broke the balloon.

Sentence (8) can also be summarized by

(11) The tree caused the balloon to break.

In this case the following rule was applied:

S2: Summary(CAUSE[X,Y])→Instrument(X) caused (Y).

Rules S1 and S2 together provide an account of how language summarizes experience as well as how one linguistic string summarizes another.

The third and fourth rules for summarizing CAUSEs are very similar and very common, but they are not as powerful and do not summarize as concisely.

S3: Summary(CAUSE[X,Y] → (X) {0|and|so|caused} (Y)

Thus the proposition underlying (8) can be stated in any of four equivalent ways, and appears originally in form (c):

(12a) The wind carried the balloon into a branch of a tree. It broke.

(12b) The wind carried the balloon into a branch of a tree and it broke.

(12c) The wind carried the balloon into a branch of a tree so it broke.

(12d) The wind carried the balloon into a branch of a tree causing it to break.

The fourth summarization rule is:

S4: Summary(CAUSE[X,Y] → (Y) {because|when|after} (X)

which puts the effect before the cause in three forms:

(13a) The balloon broke because the wind carried it into a tree.

(13b) The balloon broke when the wind carried it into a tree.

- (13c) The balloon broke after the wind carried it into a tree.
- (13d) The countryman's son died because a snake bit him.
- (13e) The countryman's son died when a snake bit him.
- (13f) The countryman's son died after a snake bit him.

Although our applications of rules S3 and S4 may not have seemed to be very short summaries of (6) and (8), it should be pointed out that the four events alluded to in (6) and (8) are already summaries of the underlying events under discussion.

C. Summarizing INITIATIONS

The highest node in an episode is an INITIATE relationship between an Event and a Reaction to that event. Thus the summarization of an episode is the summarization of an INITIATE relationship. I have developed three basic formula for the summarization of INITIATE relationships:

S5: Summary(INITIATE[X,Y]) -> (X)
 {0 | and | caused | so} (Y)

S6: Summary(INITIATE[X,Y]) -> (Y)
 {because | when | after} (X)

S7: Summary(INITIATE[X,Y]) -> (Y)

Rules S5 and S6 are obvious analogs to Rules S3 and S4 and similar examples to those illustrated in (14) could be given. Rule S7 illustrates the emphasis on the second argument, the reaction, of the INITIATE relationship. Thus Episodes can occasionally be summarized by a summary of the Reaction alone with no reference to the INITIATING Event. Summarization of this sort will be illustrated somewhat later.

D. Summarizing ALLOWs

ALLOW(Event,Event) is a rather weak causal relationship with a strong emphasis on the ALLOWed Event. Thus summarizations of the ALLOW relationship do not include the explicit reference to causal terms and are very frequently summarized with reference only to the ALLOWed Event. We thus obtain the following summarization rules:

S8: Summary(ALLOW[X,Y] -> (X) { $\emptyset$ | and} (Y)

S9: Summary(ALLOW[X,Y]) -> (Y) after (X)

S10: Summary(ALLOW[X,Y]) -> (Y)

In the serpent story, I analyzed the relationship between the snake turning and the snakes killing the boy to be an ALLOW relationship. This relationship can be summarized then, in the following ways:

- (14a) The snake turned toward the boy. He killed him.
- (14b) The snake turned toward the boy and killed him.
- (14c) The snake killed the boy after he turned toward him.
- (14d) The snake killed the boy.

E. Summarizing MOTIVATIONS

MOTIVATE is a relationship that holds between a thought and a response to that thought. It appears that MOTIVATES are differentially summarized depending on the thought in question and depending on the degree to which the response to the thought adequately carried out the intentions of the thought. I thus formulated four summarization rules for MOTIVATE--one rule which applies generally, one for responses to Desires and two for responses to Plans.

S10': Summary(MOTIVATE[thought,response]) -> (response)

Rule 10 expresses the general tendency in summaries to delete mention of the thought and assume that the MOTIVATION of a response is fairly well predicted by the nature of the response. Thus in the Margie story there was no need to mention Margie's sadness at the breaking of her balloon because it was more or less implied by her crying. This is very common and perhaps in the bulk of the cases the "thoughts" are not explicitly referred to.

**S11: Summary(MOTIVATE[Desire,response]) -> Desirer
{got | managed to get | tried to get} (Desire) by
(Response)**

In this case the terms "got" or "managed to get" are used just in case the response in fact accomplishes the Desire. The phrase "tried to get" is used in the case the response failed to accomplish the Desire. Furthermore, the "by phrase" of the rule is optional and may be deleted. Thus we obtain summaries of portions of the serpent story such as:

- (15) The man tried to make peace by bringing the snake food.
- (16) . . . the snake got revenge by killing some of his cattle.
- (17) . . . a farmer tried to make peace.

In the case where the thought is a Plan, a similar, but slightly different, rule seems to apply.

**S12: Summary(MOTIVATE[Plan,response]) -> Planner
{tried to do | did | managed to do} (Plan)
{so | but | and | Ø} Consequent(response)**

In this case again if the plan was unsuccessful, we obtain the phrase "tried to do plan" and "but Consequence" whereas if the plan was successful, we would be more likely to obtain "did plan", "managed to do plan" and "so consequence". One example from the serpent story is:

- (18) The man managed to cut off part of the snake's tail.

(Note in this example the Consequence has been lexicalized along with the action in the verb.)

Finally, we have

S13: Summary(MOTIVATE[Plan,response]) ->
 Consequent(response) {when | because | after | by}
 Planner {tried to do | managed to do | did} (Plan)

Here we obtain examples such as:

- (19) A dog lost his meat in a stream when he tried to grab its reflection in the water.
 (20) A greedy dog lost his meat by trying to get the meat held by his reflection.

F. Summarizing ANDs and THENs

In addition to summarizing the above four semantic relations, the two sequencing relationships are also summarized. No formal rules will be attempted here, but it can be said that ANDs and THENs are both summarized primarily by deletion. Two rules seem to determine which get deleted:

(1) Those arguments which dominate the fewest nodes seem to get deleted first.

(2) In a temporally sequenced set of arguments the first and the last are the last to be deleted. The last argument in the sequence is usually the very last to be deleted.

G. Generation of Summaries

I will proceed, in this section, by showing how the rules developed in the prior section can be used to account for the summaries we have observed. I will begin with the

Margie story. Consider, first, the summary given by my first subject for this story.

(21) Margie cried when the wind broke her balloon.

I will now consider how an approximation to the summary (21) might be derived according to our summarization rules. With the story parsed according to the grammar, the summary for the entire story becomes just a summary of the episode through the deletion of the setting by rule S10. Sequential application of rules S6, S1, and S10 generate the following form of summary:

(22) Summary(reaction) *when* Agent(event1) *caused*
 Summary(event2)

We can then insert the propositional and nominal referents for the two summaries and the agent. Then if we delete the second of the conjoined arguments in "Margie cried and cried", and make the proposition embedded under cause into an infinitive, we obtain the summary:

(23) Margie cried when the wind caused her balloon to burst.

If we in addition realize that "burst" is "to become broken in a certain manner" and that "cause to become broken" is the same as "break" we can derive the same summary given by our subject.

Now consider the derivation of the summary for the serpent story:

(24) A farmer cut the tail off of a snake after it had killed his son. After the snake attacked the farmer's cattle, the farmer asked for a truce, but the snake refused.

The closest derivation of (24) is:

(25) A farmer cut off a serpent's tail after it killed his son. After the serpent stung the farmer's cattle, the farmer tried to make peace, but the snake refused.

These examples are designed to illustrate how the summarization rules can interact with the semantic and syntactic rules to produce the kinds of summaries we observe from our subjects. Obviously, a more detailed analysis is required to determine whether these rules will account for the summaries of many people over many stories. It does seem, however, that these rules are sufficient to account for the most obvious aspects of people's summarization over a number of stories.

V. CONCLUSION

The outline presented above is obviously only a tentative beginning of a theory of the structure and summarization of stories. The summarization rules are clearly incomplete and a good deal more work is required to apply them to more complex cases. Furthermore, the grammar has difficulty handling more complex multi-protagonist stories. Nevertheless, the results to date have been encouraging and generalizations to the rules listed above are currently under development. Moreover, a number of attempts to use these rules (and generalizations of these rules) as tools for the analysis of various kinds of experimental data are currently being made. For example, preliminary results indicate that "more structured stories" are easier to recall than less structured stories. An attempt is being made, within the grammar, to define the notion of degree of structure in a way consonant with the observed results. In a forthcoming paper I will report on a more detailed analysis of summarizations collected from additional subjects on a wider range of story types. A version of the grammar is under development which will account for the flow of a conversation and later summarization of that conversation. The structures generated by the grammar are being used to propose answers to *why-questions*. A multi-protagonist version of the grammar is under development. Although these results are not all in yet, it now appears that the basic character of the grammar presented above will allow us to account for a substantial range of phenomena related to the higher order structures found in stories.

Before concluding, a comment about the relationship of

this approach to that developed by other workers is in order. In this volume, the work most closely related to this is to be found in Chapters 9 and 10 by Schank and by Abelson. I see both of those lines of attack as generally consonant with my own. The work presented here differs from Schank's primarily in that I suggest global rules and attempt to describe the global structure of a story. Schank, on the other hand, emphasizes the "bottom-up" need to integrate each sentence into a paragraph. Using his methods, the first level relationships among the sentences become clear, but no systematic higher order structure is presupposed. It is my suspicion that any automatic "story parser" would require the "top-down" global structures that I propose, but would to a large degree discover them, in any given story, by the procedures developed by Schank.

With respect to work outside this volume it should be added that the approach developed here is, to some degree, designed to be a systematization of Propp's (1968) analysis of Russian folk tales. Although it hasn't been shown here, the rules presented above are designed to capture the relationships among the structures developed by Propp. In that light, this work should be compared to that of Colby (1973) on Eskimo folktales.

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